FORM 2

THE PATENTS ACT, 1970

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The Patent Rules, 2003

COMPLETE SPECIFICATION

(See sections 10 & rule 13)

1. TITLE OF THE INVENTION

SYSTEM AND METHOD FOR THREE-DIMENSIONAL ULTRASOUND IMAGING

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3. PREAMBLE TO THE DESCRIPTION

COMPLETE SPECIFICATION

The following specification particularly describes the invention and the manner in which it is to be performed.

TECHNICAL FIELD

[0001] The present disclosure generally relates to the field of medical imaging technology. In particular, the present disclosure relates to a system and a method for three-dimensional ultrasound imaging using a pivoted acoustic bed and at least one stationary acoustic reflector.

BACKGROUND

[0002] Three-dimensional ultrasound imaging is widely used in medical diagnostic imaging to visualize anatomical structures with improved spatial context compared to two-dimensional B-mode imaging. In many workflows, a three-dimensional image volume is generated by acquiring a plurality of two-dimensional B-mode frames while an ultrasound probe is moved through a controlled trajectory, and then reconstructing the plurality of two-dimensional B-mode frames into a volumetric representation.

[0003] Conventional approaches for generating a three-dimensional image volume using a two-dimensional ultrasound probe typically rely on external tracking mechanisms to determine probe position and/or orientation during acquisition. Such external tracking mechanisms may include optical tracking systems, electromagnetic tracking systems, inertial sensing units, or mechanical encoders. However, these tracking mechanisms often increase system cost and complexity, may require line-of-sight or interference-free environments, and may be sensitive to drift, occlusion, or setup variability, thereby reducing reliability in routine clinical use.

[0004] In addition, existing tracking-based solutions may be cumbersome for portable or wearable implementations. The need for external trackers, frequent calibration routines, and additional attachments may limit ease of deployment in bedside monitoring, point-of-care diagnostics, and longer-duration imaging scenarios where compactness and repeatable operation are preferred.

[0005] Therefore, there is a need for an improved approach that overcomes at least the above-mentioned drawbacks.

OBJECTS OF THE PRESENT DISCLOSURE

[0006] A general object of the present disclosure is to provide a system and a method for three-dimensional ultrasound imaging using a pivoted acoustic bed and an ultrasound probe configured to acquire multiple two-dimensional B-mode frames.

[0007] Another object of the present disclosure is to generate, using one or more stationary acoustic reflectors integrated into the pivoted acoustic bed, a reflector signature in each two-dimensional B-mode frame to enable intrinsic positional coding.

[0008] Another object of the present disclosure is to determine a slice position corresponding to each two-dimensional B-mode frame by applying closed-form geometric mapping based on a depth of the reflector signature and stored calibration parameters.

[0009] Another object of the present disclosure is to reconstruct a three-dimensional image volume by ordering two-dimensional B-mode frames according to the slice positions and processing the ordered frames using a beamforming-based reconstruction technique.

[0010] Another object of the present disclosure is to reduce reliance on external probe tracking mechanisms by using a reflector-signature-based approach for slice position determination in portable, point-of-care, or wearable ultrasound imaging scenarios.

SUMMARY

[0011] Aspects of the present disclosure pertain to the field of medical imaging technology. In particular, the present disclosure relates to a system and a method for three-dimensional ultrasound imaging using a pivoted acoustic bed and at least one stationary acoustic reflector.

[0012] An aspect of the present disclosure pertains to a system for three-dimensional ultrasound imaging. The system includes a pivoted acoustic bed. The system further includes an ultrasound probe pivotally mounted on the pivoted acoustic bed and configured to acquire two-dimensional B-mode frames. The system further includes one or more stationary acoustic reflectors integrated into the pivoted acoustic bed and configured to generate, in each of the two-dimensional B-mode frames, a reflector signature including a high-energy line at a depth. The system also includes a processing unit operatively coupled to the ultrasound probe. The processing unit includes one or more processors and a memory storing instructions. When executed, the instructions cause the processing unit to detect the reflector signature in each of the two-dimensional B-mode frames. The instructions further cause the processing unit to determine, based on the depth of the reflector signature and at least one stored calibration parameter including a reflector angle and at least one offset, a slice position corresponding to each two-dimensional B-mode frame using closed-form geometric mapping. The instructions further cause the processing unit to reconstruct, based on the slice positions and the two-dimensional B-mode frames, a three-dimensional image volume.

[0013] In one embodiment, the pivoted acoustic bed is configured to pivot the ultrasound probe about at least one axis to obtain multiple two-dimensional B-mode frames at different angular positions.

[0014] In one embodiment, the one or more stationary acoustic reflectors is configured such that the reflector signature is repeatably detectable across a range of probe angular positions to provide robust positional coding.

[0015] In one embodiment, the processing unit is configured to perform a one-time calibration procedure to determine and store the reflector angle and the at least one offset as the at least one stored calibration parameter.

[0016] In one embodiment, detecting the reflector signature includes identifying the high-energy line in each two-dimensional B-mode frame. Detecting the reflector signature further includes extracting a pixel-depth value corresponding to the high-energy line.

[0017] In one embodiment, the processing unit is configured to order the two-dimensional B-mode frames according to the slice positions. The processing unit is further configured to reconstruct the three-dimensional image volume using a beamforming-based reconstruction technique.

[0018] Another aspect of the present disclosure pertains to a method for three-dimensional ultrasound imaging. The method includes acquiring, by an ultrasound probe pivotally mounted on a pivoted acoustic bed, a plurality of two-dimensional B-mode frames. The method further includes generating, by one or more stationary acoustic reflectors integrated into the pivoted acoustic bed, in each of the two-dimensional B-mode frames, a reflector signature comprising a high-energy line at a depth. The method further includes detecting, by a processing unit operatively coupled to the ultrasound probe, the reflector signature in each of the two-dimensional B-mode frames. The method further includes determining, by the processing unit, based on the depth of the reflector signature and at least one stored calibration parameter comprising a reflector angle and at least one offset, a slice position corresponding to each two-dimensional B-mode frame using closed-form geometric mapping. The method further includes reconstructing, by the processing unit, based on the slice positions and the two-dimensional B-mode frames, a three-dimensional image volume.

BRIEF DESCRIPTION OF THE DRAWINGS

[0019] The accompanying drawings illustrate exemplary embodiments of the present disclosure and, together with the description, serve to explain the principles of the present disclosure.

[0020] FIG. 1 illustrates an exemplary arrangement of a system (100) for three-dimensional ultrasound imaging using an ultrasound probe (102), in accordance with an embodiment of the present disclosure.

[0021] FIG. 2 illustrates an exemplary arrangement of a processing unit (110) operatively coupled to the ultrasound probe (102), in accordance with an embodiment of the present disclosure.

[0022] FIG. 3A illustrates an exemplary structural arrangement of a pivoted acoustic bed (116), in accordance with an embodiment of the present disclosure.

[0023] FIG. 3B illustrates an exemplary arrangement of an ultrasound probe (102) mounted on a pivoted acoustic bed (116), in accordance with an embodiment of the present disclosure.

[0024] FIG. 4 illustrates an exemplary geometric representation for determining slice position of an ultrasound probe (102) moved along a curved scan track (107), in accordance with an embodiment of the present disclosure.

[0025] FIG. 5 illustrates an exemplary block diagram of the processing unit (110), in accordance with an embodiment of the present disclosure.

[0026] FIG. 6 illustrates an exemplary flow chart of a method (600) for three-dimensional (3D) ultrasound imaging, in accordance with an embodiment of the present disclosure.

DETAILED DESCRIPTION

[0027] In the following description, for the purposes of explanation, various specific details are set forth in order to provide a thorough understanding of embodiments of the present disclosure. It will be apparent, however, that embodiments of the present disclosure may be practiced without these specific details. Several features described hereafter can each be used independently of one another or with any combination of other features. An individual feature may not address all of the problems discussed above or might address only some of the problems discussed above. Some of the problems discussed above might not be fully addressed by any of the features described herein.

[0028] The ensuing description provides exemplary embodiments only and is not intended to limit the scope, applicability, or configuration of the disclosure. Rather, the ensuing description of the exemplary embodiments will provide those skilled in the art with an enabling description for implementing an exemplary embodiment. It should be understood that various changes may be made in the function and arrangement of elements without departing from the spirit and scope of the disclosure as set forth.

[0029] The following is a detailed description of embodiments of the disclosure depicted in the accompanying drawings. The embodiments are in such detail as to clearly communicate the disclosure. If the specification states a component or feature “may”, “can”, “could”, or “might” be included or have a characteristic, that particular component or feature is not required to be included or have the characteristic.

[0030] As used in the description herein and throughout the claims that follow, the meaning of “a,” “an,” and “the” includes plural reference unless the context clearly dictates otherwise. Also, as used in the description herein, the meaning of “in” includes “in” and “on” unless the context clearly dictates otherwise.

[0031] Conventional three-dimensional ultrasound imaging using a two-dimensional ultrasound probe generally requires an external tracking mechanism to determine probe position and orientation during acquisition. Such tracking mechanisms increase cost and operational complexity, may suffer from drift, occlusion, or environmental interference, and often require repeated setup and calibration. These constraints make it difficult to deploy reliable three-dimensional imaging in portable, point-of-care, bedside, and wearable scenarios where compactness and repeatable operation are preferred.

[0032] The disclosed approach provides a pivoted acoustic bed that supports an ultrasound probe and enables acquisition of multiple two-dimensional B-mode frames at different angular positions. A stationary acoustic reflector integrated into the pivoted acoustic bed generates, in each two-dimensional B-mode frame, a reflector signature comprising a high-energy line at a depth. A processing unit detects the reflector signature, determines a slice position for each two-dimensional B-mode frame using closed-form geometric mapping based on the reflector-signature depth and stored calibration parameters including a reflector angle and at least one offset, and reconstructs a three-dimensional image volume from the slice positions and the two-dimensional B-mode frames.

[0033] By deriving slice positions from a reflector signature embedded within the acquired two-dimensional B-mode frames, the disclosed approach reduces reliance on external tracking mechanisms while maintaining consistent slice ordering for volumetric reconstruction. The use of closed-form geometric mapping and stored calibration parameters enables efficient and repeatable slice position determination across a range of probe angular positions. As a result, the approach supports practical three-dimensional ultrasound imaging with reduced setup burden and improved suitability for portable and wearable imaging configurations.

[0034] The present disclosure pertains to the field of medical imaging technology. In particular, the present disclosure relates to a system and a method for three-dimensional ultrasound imaging using a pivoted acoustic bed and at least one stationary acoustic reflector.

[0035] Various embodiments with respect to the present disclosure will be explained in detail with reference to FIGs. 1-6.

[0036] FIG. 1 illustrates an exemplary arrangement of a system (100) for three-dimensional ultrasound imaging using an ultrasound probe (102), in accordance with one or more embodiments of the present disclosure.

[0037] The ultrasound probe (102) may be configured to acquire at least one two-dimensional (2D) B-mode image (104). As illustrated, the 2D B-mode image (104) may be defined in an imaging plane having an in-plane axis X and a depth axis Z.

[0038] The ultrasound probe (102) may be operatively supported by a pivotable assembly (106). The pivotable assembly (106) may guide motion of the ultrasound probe (102) along a curved scan track (107), such that the ultrasound probe (102) may acquire a plurality of 2D ultrasound frames (108) of a target anatomy (as shown in FIG. 1 slices (111) of target anatomy) at different angular positions along the curved scan track (107).

[0039] The plurality of 2D ultrasound frames (108) may each include a reflector signature (109). The reflector signature (109) may correspond to a high-energy line at a depth within a respective 2D ultrasound frame (108). In the illustrated example, the reflector signature (109) is shown as being present across the plurality of 2D ultrasound frames (108).

[0040] The plurality of 2D ultrasound frames (108), including the reflector signature (109), may be provided to a processing unit (110). The processing unit (110) may include one or more processors and a memory for processing the plurality of 2D ultrasound frames (108). In one embodiment, the processing unit (110) may be implemented using a computing device, such as a laptop computer, a workstation, or an embedded computing platform, but not limited thereto.

[0041] The processing unit (110) may generate a three-dimensional volumetric reconstruction (112) of a target anatomy based on the plurality of 2D ultrasound frames (108). As further illustrated, a stationary acoustic reflector (114) may be arranged relative to the three-dimensional volumetric reconstruction (112) and may be associated with generation of the reflector signature (109) in the plurality of 2D ultrasound frames (108).

[0042] FIG. 2 illustrates an exemplary arrangement of a processing unit (110) operatively coupled to the ultrasound probe (102), in accordance with one or more embodiments of the present disclosure.

[0043] As illustrated, the ultrasound probe (102) may provide acquired ultrasound data to the processing unit (110) via a communication link. The communication link may be implemented using a wired interface or a wireless interface, but not limited thereto. The processing unit (110) may include one or more processor(s) (204) and a memory (206). The one or more processor(s) (204) may be configured to execute instructions stored in the memory (206) for processing data received from the ultrasound probe (102). The memory (206) may store, for example, executable instructions, calibration parameters, and intermediate data associated with processing of the ultrasound data, but not limited thereto.

[0044] In one embodiment, a calibration parameter may include one or more stored values used by the processing unit (110) to relate a measured depth of the reflector signature (109) in a 2D B-mode frame (108) to a corresponding slice position using closed-form geometric mapping. The calibration parameter may include a reflector angle that characterizes an orientation of the stationary acoustic reflector (114) relative to an imaging geometry of the ultrasound probe (102). The calibration parameter may further include at least one offset that represents a positional shift between a reference point associated with the ultrasound probe (102) and a reference point associated with the stationary acoustic reflector (114). In one embodiment, the at least one offset may correspond to an axial offset along a depth direction, a lateral offset along an in-plane direction, or an origin offset associated with a curved scan track (107), but not limited thereto. In one embodiment, the processing unit (110) may determine and store the calibration parameter using a one-time calibration procedure, such that the stored calibration parameter is later applied during detection of the reflector signature (109) and slice position determination for volumetric reconstruction.

[0045] FIG. 3A illustrates an exemplary structural arrangement of a pivoted acoustic bed (116), in accordance with one or more embodiments of the present disclosure.

[0046] The pivoted acoustic bed (116) may include a pivotable probe holder (118) configured to support an ultrasound probe (not shown) during imaging. The pivotable probe holder (118) may be arranged to move relative to the pivoted acoustic bed (116) along a curved scan track (107).

[0047] The pivoted acoustic bed (116) may further include one or more stationary acoustic reflector(s) (114A-114B) arranged within the pivoted acoustic bed (116). As illustrated, a first stationary acoustic reflector (114A) and a second stationary acoustic reflector (114B) may be positioned on opposing sides of the pivotable probe holder (118) along the curved scan track (107).

[0048] The pivotable probe holder (118) may be configured to pivot about at least one axis (120). In one embodiment, pivoting about the at least one axis (120) may enable acquisition of multiple 2D B-mode frames at different angular positions along the curved scan track (107).

[0049] In one embodiment, the placement, positioning, count, and type of the one or more stationary acoustic reflector(s) (114A-114B) illustrated in FIG. 3A are provided for purposes of explanation and illustration. The one or more stationary acoustic reflector(s) (114A-114B) may be arranged in other positions, in other quantities, and/or using other reflector geometries or materials, but not limited thereto, provided that the one or more stationary acoustic reflector(s) (114A-114B) are configured to generate a reflector signature in acquired 2D B-mode frames for determining slice position. In one embodiment, the pivoted acoustic bed (116) may include a single stationary acoustic reflector stripe within the acoustic exposure of the ultrasound probe (102).

[0050] FIG. 3B illustrates an exemplary arrangement of an ultrasound probe (102) mounted on a pivoted acoustic bed (116), in accordance with one or more embodiments of the present disclosure.

[0051] As illustrated, the ultrasound probe (102) may be supported by a pivotable probe holder (118) disposed on the pivoted acoustic bed (116). The pivotable probe holder (118) may be configured to guide movement of the ultrasound probe (102) along a curved scan track (107) during acquisition of ultrasound data.

[0052] The pivoted acoustic bed (116) may include a first stationary acoustic reflector (114A) and a second stationary acoustic reflector (114B) arranged on opposing sides of the ultrasound probe (102). The first stationary acoustic reflector (114A) and the second stationary acoustic reflector (114B) may be positioned relative to the curved scan track (107) such that, during movement of the ultrasound probe (102), the first stationary acoustic reflector (114A) and the second stationary acoustic reflector (114B) may generate reflector signatures in acquired 2D B-mode frames.

[0053] The ultrasound probe (102) may be configured to pivot about an axis (120) while moving along the curved scan track (107). In one embodiment, pivoting about the axis (120) may enable acquisition of multiple 2D B-mode frames at different angular positions along the curved scan track (107).

[0054] FIG. 4 illustrates an exemplary geometric representation (sub-figure (a)) for determining slice position of an ultrasound probe (102) moved along a curved scan track (107), in accordance with one or more embodiments of the present disclosure.

[0055] As illustrated in an upper portion of FIG. 4, the ultrasound probe (102) may occupy a plurality of positions along the curved scan track (107), including a first position S_B, an intermediate position S_N, and a second position S_A. In one embodiment, the first position S_B and the second position S_A may define an arc span along the curved scan track (107) having a length L_{AB}.

[0056] A stationary acoustic reflector (114) is illustrated relative to the curved scan track (107). The stationary acoustic reflector (114) may be oriented at an angle θ relative to a reference direction, and may be associated with a depth-varying reflector response in frames acquired by the ultrasound probe (102) as the ultrasound probe (102) traverses the curved scan track (107).

[0057] In one embodiment, a depth measurement corresponding to the reflector response may vary as a function of the position of the ultrasound probe (102) along the curved scan track (107). For instance, the depth measurement may correspond to d_B when the ultrasound probe (102) is at the first position S_B, may correspond to d_N when the ultrasound probe (102) is at the intermediate position S_N, and may correspond to d_A when the ultrasound probe (102) is at the second position S_A. The schematic further illustrates geometric parameters including a height H, a depth span D, and a reflector-response thickness T_R, which may be used to relate the measured depth to a slice position along the curved scan track (107).

[0058] A lower portion of FIG. 4 illustrates exemplary 2D B-mode images (104) corresponding to the first position S_B (sub-figure (b)), the intermediate position S_N (sub-figure (c)), and the second position S_A (sub-figure (d)). As illustrated, each 2D B-mode image (104) may include a reflector response that is observable at the respective depth (d_B, d_N, or d_A), and the reflector-response thickness T_R may be measurable in the 2D B-mode images (104) at least for some probe positions.

[0059] In one embodiment, the processing unit (110) may implement the closed-form geometric mapping by converting a measured depth of the reflector signature (109) in a 2D B-mode frame (108) into a slice position along the curved scan track (107) using stored calibration parameters. With reference to FIG. 4, the measured depth may be represented as a depth value such as d_B, d_N, or d_A depending on a current probe position along the curved scan track (107). The stored calibration parameters may include the reflector angle (θ) and at least one offset. The at least one offset may define, for example, a reference depth shift or a reference origin shift between the ultrasound probe (102) and the stationary acoustic reflector (114). In one exemplary implementation, the processing unit (110) may first apply the at least one offset to obtain an adjusted depth value d′. The processing unit (110) may then compute a corresponding slice position parameter by applying a closed-form relationship that uses the reflector angle (θ) and the adjusted depth value d. For example, when the stationary acoustic reflector (114) is treated as a straight-line feature in the geometric model of FIG. 4, the processing unit (110) may compute a proportional distance along the scan span L_{AB} or a corresponding angular index based on a ratio between the adjusted depth value d and a depth span such as H or D, and may use the reflector angle (θ) to account for the slanted orientation of the stationary acoustic reflector (114), but not limited thereto. In one embodiment, the processing unit (110) may then order the 2D B-mode frames (108) using the computed slice positions and reconstruct the 3D image volume (112).

[0060] In one embodiment, the processing unit (110) may perform a one-time calibration procedure to determine and store the reflector angle (θ) and the at least one offset used as the stored calibration parameter. The one-time calibration procedure may include positioning the ultrasound probe (102) at a first known angular position along the curved scan track (107), acquiring at least one 2D B-mode frame (108), and detecting the reflector signature (109) to obtain a first reference depth. The one-time calibration procedure may further include positioning the ultrasound probe (102) at a second known angular position along the curved scan track (107), acquiring at least one additional 2D B-mode frame (108), and detecting the reflector signature (109) to obtain a second reference depth. Based on the first reference depth and the second reference depth and the known separation between the first known angular position and the second known angular position, the processing unit (110) may determine the reflector angle (θ) associated with the stationary acoustic reflector (114). The processing unit (110) may further determine the at least one offset that aligns the measured depths to a reference coordinate system used for the closed-form geometric mapping. The processing unit (110) may store the reflector angle (θ) and the at least one offset in the memory (204) for subsequent slice position determination during volumetric reconstruction, but not limited thereto.

[0061] FIG. 5 illustrates an exemplary block diagram of the processing unit (110), in accordance with one or more embodiments of the present disclosure. The processing unit (110) may include one or more processor(s) (204), a main memory (206), a read-only memory (506), a mass storage device (508), one or more communication port(s) (510), a bus (512), an external storage device (514), and a user interface (516). The elements of the processing unit (110) may be operatively coupled to one another by the bus (512).

[0062] The one or more processor(s) (204) may be configured to execute instructions stored in at least one of the main memory (206), the read-only memory (506), the mass storage device (508), or the external storage device (514) to perform processing functions associated with the present disclosure. In one embodiment, the one or more processor(s) (204) may include one or more microprocessors, microcontrollers, digital signal processors, system-on-chip devices, or application-specific integrated circuits, but not limited to thereto.

[0063] The main memory (206) may include volatile memory configured to store runtime data, intermediate results, and executable instructions during operation of the processing unit (110). The read-only memory (506) may include non-volatile memory configured to store boot instructions, initialization routines, or configuration parameters for operation of the processing unit (110). The mass storage device (508) may provide persistent storage for software modules, datasets, logs, and/or reconstructed output data, but not limited to thereto.

[0064] The one or more communication port(s) (510) may enable bidirectional communication between the processing unit (110) and one or more external devices. In one embodiment, the one or more communication port(s) (510) may support one or more wired interfaces or wireless interfaces, but not limited thereto. The external storage device (514) may be operatively coupled to the bus (512) and may enable import and/or export of data, software, or configuration information, such as via removable media or network-access storage, but not limited to thereto.

[0065] The user interface (516) may enable local interaction with the processing unit (110). In one embodiment, the user interface (516) may include one or more displays and one or more input devices to support visualization of outputs and entry of user inputs, but not limited thereto.

[0066] The configuration shown in FIG. 5 is illustrative. In one embodiment, one or more components of the processing unit (110) may be combined, omitted, or implemented using distributed computing resources without departing from the scope of the present disclosure.

[0067] In one embodiment, the system (100) for three-dimensional (3D) ultrasound imaging may include a pivoted acoustic bed (116) and an ultrasound probe (102) pivotally mounted on the pivoted acoustic bed (116). The ultrasound probe (102) may be configured to acquire a plurality of two-dimensional (2D) B-mode frames. In one embodiment, the pivoted acoustic bed (116) may include a pivotable probe holder (118) that supports the ultrasound probe (102) and guides motion of the ultrasound probe (102) along a curved scan track (107) to obtain the plurality of 2D B-mode frames at different angular positions.

[0068] In one embodiment, the system (100) may include a stationary acoustic reflector (114) integrated into the pivoted acoustic bed (116). The stationary acoustic reflector (114) may be configured to generate, in each of the plurality of 2D B-mode frames acquired by the ultrasound probe (102), a reflector signature (109) comprising a high-energy line at a depth. The reflector signature (109) may be detectable as a consistent feature within the plurality of 2D B-mode frames.

[0069] In one embodiment, the system (100) may include a processing unit (110) operatively coupled to the ultrasound probe (102). The processing unit (110) may include one or more processors and a memory storing instructions that, when executed, cause the processing unit (110) to detect the reflector signature (109) in each of the plurality of 2D B-mode frames. The processing unit (110) may further determine, based on the depth of the reflector signature (109) and at least one stored calibration parameter comprising a reflector angle and at least one offset, a slice position corresponding to each 2D B-mode frame using closed-form geometric mapping. The processing unit (110) may then reconstruct, based on the slice positions and the plurality of 2D B-mode frames, a 3D image volume (112).

[0070] In one embodiment, the system (100) may include the pivoted acoustic bed (116) configured to pivot the ultrasound probe (102) about at least one axis (120). Pivoting of the ultrasound probe (102) about the at least one axis (120) may enable the ultrasound probe (102) to obtain multiple 2D B-mode frames at different angular positions along the curved scan track (107).

[0071] In one embodiment, the system (100) may include the stationary acoustic reflector (114) configured such that the reflector signature (109) is repeatably detectable across a range of angular positions of the ultrasound probe (102). In one embodiment, repeatable detectability of the reflector signature (109) across the range of angular positions may provide robust positional coding for determining a slice position corresponding to a respective 2D B-mode frame.

[0072] In one embodiment, the processing unit (110) of the system (100) may be configured to perform a one-time calibration procedure. The one-time calibration procedure may determine a reflector angle and at least one offset associated with the stationary acoustic reflector (114). The processing unit (110) may store the reflector angle and the at least one offset as the at least one stored calibration parameter.

[0073] In one embodiment, detecting the reflector signature (109) may include identifying the high-energy line in each 2D B-mode frame acquired by the ultrasound probe (102). Detecting the reflector signature (109) may further include extracting a pixel-depth value corresponding to the high-energy line within the respective 2D B-mode frame.

[0074] In one embodiment, the processing unit (110) of the system (100) may be configured to order the 2D B-mode frames according to the slice positions determined for the respective 2D B-mode frames. The processing unit (110) may further reconstruct the 3D image volume (112) based on the ordered 2D B-mode frames using a beamforming-based reconstruction technique.

[0075] Referring now to FIG. 6, an exemplary flow chart of a method (600) for three-dimensional (3D) ultrasound imaging is illustrated, in accordance with one or more embodiments of the present disclosure. In one embodiment, the method (600) may be implemented by the system (100) including the ultrasound probe (102), the pivoted acoustic bed (116), and the processing unit (110).

[0076] At step (602), the method (600) may include acquiring, by the ultrasound probe (102) pivotally mounted on the pivoted acoustic bed (116), a plurality of two-dimensional (2D) B-mode frames. In one embodiment, the pivoted acoustic bed (116) may pivot the ultrasound probe (102) about at least one axis (120) to obtain the plurality of 2D B-mode frames at different angular positions.

[0077] At step (604), the method (600) may include generating, by a stationary acoustic reflector (114) integrated into the pivoted acoustic bed (116), in each of the 2D B-mode frames, a reflector signature (109) comprising a high-energy line at a depth.

[0078] At step (606), the method (600) may include detecting, by the processing unit (110) operatively coupled to the ultrasound probe (102), the reflector signature (109) in each of the 2D B-mode frames. In one embodiment, detecting the reflector signature (109) may include identifying the high-energy line and extracting a pixel-depth value corresponding to the high-energy line.

[0079] At step (608), the method (600) may include determining, by the processing unit (110), based on the depth of the reflector signature (109) and at least one stored calibration parameter comprising a reflector angle and at least one offset, a slice position corresponding to each 2D B-mode frame using closed-form geometric mapping. In one embodiment, the at least one stored calibration parameter may be obtained using a one-time calibration procedure performed by the processing unit (110).

[0080] At step (610), the method (600) may include reconstructing, by the processing unit (110), based on the slice positions and the 2D B-mode frames, a 3D image volume (112). In one embodiment, the processing unit (110) may order the 2D B-mode frames according to the slice positions and reconstruct the 3D image volume (112) using a beamforming-based reconstruction technique.

[0081] It may be appreciated that one or more steps of the method (600) may be performed in a different order, combined, omitted, or supplemented with additional steps, without departing from the scope of the present disclosure.

[0082] While the foregoing describes example embodiments, other and further embodiments may be devised without departing from the scope defined by the claims, and the embodiments, versions, and examples are non-limiting and are provided solely to enable a person having ordinary skill in the art to make and use the present disclosure in view of their general knowledge.

ADVANTAGES OF THE PRESENT DISCLOSURE

[0083] The present disclosure provides three-dimensional ultrasound imaging using a pivoted acoustic bed and a stationary acoustic reflector, thereby enabling slice position determination from reflector signatures in acquired two-dimensional B-mode frames.

[0084] The present disclosure provides intrinsic positional coding via a reflector signature comprising a high-energy line at a depth, thereby reducing dependence on external tracking mechanisms for reconstructing a three-dimensional image volume.

[0085] The present disclosure provides closed-form geometric mapping based on stored calibration parameters, thereby enabling efficient slice position computation and reducing computational complexity associated with iterative pose estimation techniques.

[0086] The present disclosure provides a one-time calibration procedure for storing a reflector angle and at least one offset, thereby reducing recurring calibration burden and improving repeatability across imaging sessions.

[0087] The present disclosure provides ordering of two-dimensional B-mode frames according to slice positions and beamforming-based volumetric reconstruction, thereby improving volumetric consistency while maintaining compatibility with conventional ultrasound probes and scanners.

We Claim:

1. A system (100) for three-dimensional (3D) ultrasound imaging, the system (100) comprising:

a pivoted acoustic bed (116);

an ultrasound probe (102) pivotally mounted on the pivoted acoustic bed (116) and configured to acquire two-dimensional (2D) B-mode frames (108);

one or more stationary acoustic reflectors (114) integrated into the pivoted acoustic bed (116) and configured to generate, in each of the 2D B-mode frames (108), a reflector signature (109) comprising a high-energy line at a depth; and

a processing unit (110) operatively coupled to the ultrasound probe (102), the processing unit (110) comprising one or more processors (204) and a memory (206) storing instructions that, when executed, cause the processing unit (110) to:

detect the reflector signature (109) in each of the 2D B-mode frames (108);

determine, based on the depth of the reflector signature (109) and at least one stored calibration parameter comprising a reflector angle and at least one offset, a slice position corresponding to each 2D B-mode frame (108) using closed-form geometric mapping; and

reconstruct, based on the slice positions and the 2D B-mode frames (108), a 3D image volume (112).

2. The system (100) as claimed in claim 1, wherein the pivoted acoustic bed (116) is configured to pivot the ultrasound probe (102) about at least one axis (120) to obtain multiple 2D B-mode frames (108) at different angular positions.

3. The system (100) as claimed in claim 1, wherein the one or more stationary acoustic reflectors (114) is configured such that the reflector signature (109) is

repeatably detectable across a range of probe angular positions to provide robust positional coding.

4. The system (100) as claimed in claim 1, wherein the processing unit (110) is configured to perform a one-time calibration procedure to determine and store the reflector angle and the at least one offset as the at least one stored calibration parameter.

5. The system (100) as claimed in claim 1, wherein detecting the reflector signature (109) comprises identifying the high-energy line in each 2D B-mode frame (108) and extracting a pixel-depth value corresponding to the high-energy line.

6. The system (100) as claimed in claim 1, wherein the processing unit (110) is configured to order the 2D B-mode frames (108) according to the slice positions and to reconstruct the 3D image volume (112) using a beamforming-based reconstruction technique.

7. A method (600) for three-dimensional (3D) ultrasound imaging, the method (600) comprising:

acquiring (602), by an ultrasound probe (102) pivotally mounted on a pivoted acoustic bed (116), a plurality of two-dimensional (2D) B-mode frames (108);

generating (604), by one or more stationary acoustic reflectors (114) integrated into the pivoted acoustic bed (116), in each of the 2D B-mode frames (108), a reflector signature (109) comprising a high-energy line at a depth;

detecting (606), by a processing unit (110) operatively coupled to the ultrasound probe (102), the reflector signature (109) in each of the 2D B-mode frames (108);

determining (608), by the processing unit (110), based on the depth of the reflector signature (109) and at least one stored calibration parameter comprising a reflector angle and at least one offset, a slice position corresponding to each 2D B-mode frame (108) using closed-form geometric mapping; and

reconstructing (610), by the processing unit (110), based on the slice positions and the 2D B-mode frames (108), a 3D image volume (112).

For Indian Institute of Technology Palakkad

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ABSTRACT

SYSTEM AND METHOD FOR THREE-DIMENSIONAL ULTRASOUND IMAGING

The present disclosure relates to a system (100) and a method (600) for three-dimensional (3D) ultrasound imaging. The system (100) includes a pivoted acoustic bed (116), an ultrasound probe (102) pivotally mounted on the pivoted acoustic bed (116) to acquire two-dimensional (2D) B-mode frames (108), and one or more stationary acoustic reflectors (114) integrated into the pivoted acoustic bed (116) to generate, in each 2D B-mode frame (108), a reflector signature (109) including a high-energy line at a depth. A processing unit (110) including one or more processors (204) and a memory (206) detects the reflector signature (109), determines a slice position for each 2D B-mode frame (108) using closed-form geometric mapping based on the depth and at least one stored calibration parameter including a reflector angle and at least one offset, and reconstructs a 3D image volume (112).

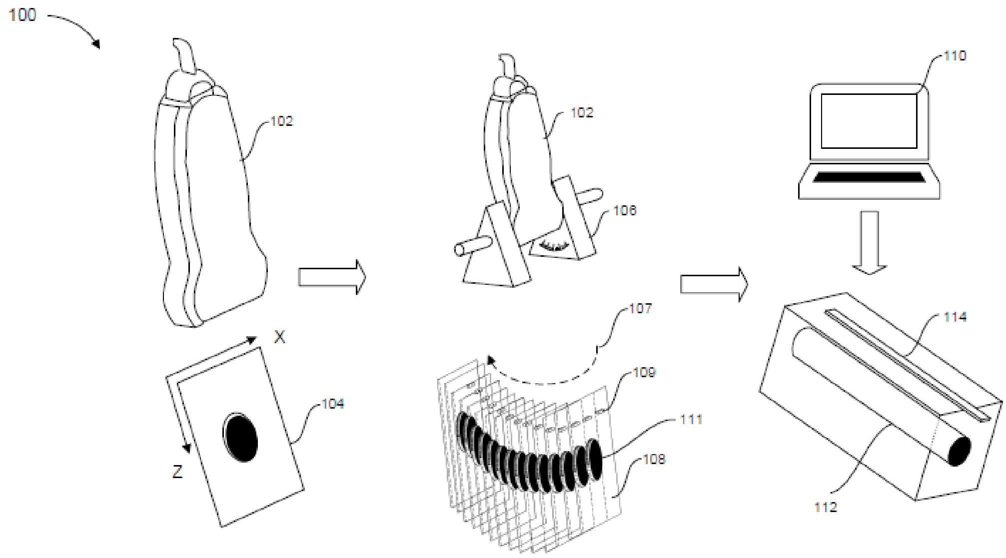

FIG. 1